

ME 300 Homework 5: Heat, Work, and the First Law

Answer the following questions neatly on a separate piece of paper or on a tablet. Written answers can be typed. Show all your work, list all your assumptions, and explain your reasoning – partial credit will only be given if you show and explain each of your steps. Submit your homework on Canvas.

1. (10 points) What does the incompressible assumption mean? Explain it in your own words.
2. (20 points) The goal of this problem is to analyze James Prescott Joule's ground-breaking experiment where he really showed the first law and wrote about it in his seminal paper, "On the Mechanical Equivalent of Heat." This problem will provide some historical context for how the first law was experimentally confirmed, and will show you calculations based on a real-life experiment that demonstrates the transfer of energy in a quantifiable way.
 - a. First, do some background research on the experiment itself and sketch it on your homework. Here are some useful resources (Definitely read/watch the first three links but you are not required to read Joule's entire paper. It's absolutely fascinating, though, so if you're interested I highly encourage you to give it a read!)
 - i. Some historical context: <http://www.aps.org/publications/apsnews/200912/physicshistory.cfm>
 - ii. An overview of the experiment: <http://twi.mpei.ac.ru/TTHB/2/KiSyShe/eng/Chapter2/2-1-Heat-Joule's-experiment-Equivalence-of-heat-and-work.html>
 - iii. A video of the experiment being recreated: <https://www.youtube.com/watch?v=MBrTDKc9YZ0>
 - iv. Joule's actual paper: <https://royalsocietypublishing.org/doi/epdf/10.1098/rstl.1850.0004>
 - b. In Joule's paper, he says that the weights he used were 58 lbs. total. The weights were lifted to 45 inches 20 times. Calculate the total change in potential energy of the weights through the experiment in Joules.
 - c. Assume water is compressible and has a specific heat of $c=4.187$ kJ/kg-K. Using the approximation that $\Delta U=Mc\Delta T$ to estimate how much the temperature changed as a result of the weights falling 20 times. Assume the mass of the water is 7.84 lbs., as stated on page 70 of Joule's paper.
 - d. Compare that temperature difference to the measurements shown in the video (or to Joule's own measurements in the paper). Why might differences arise? For hints, Joule discusses these in detail in his paper.
 - e. Using the initial temperature of the water from the video and the approximation $\Delta U=Mc\Delta T$, determine how far these weights would have to fall to boil the water.
3. (30 points) Air ($M=5$ kg) is trapped in a sealed piston system with a volume of $V_1=1$ m³ that is initially locked so that the piston can't move. The air is heated from $T_1=300$ K to $T_2=500$ K. At this point, the lock breaks, the heater goes haywire, and the piston rises in an isobaric process until the final volume of $V_3=3$ m³ is reached. Calculate the following knowing that $R=287$ J/kg-K and $c_v=714$ J/kg-K:
 - a. The pressure, temperature, and volume of the gas at all three states
 - b. The change in internal energy (units of Joules) of the gas for these two processes
 - c. The work (units of Joules) done for these two processes – was work done "by" or "on" the fluid?
 - d. The heat transfer (units of Joules) for these two processes.

4. (20 points) Initially, 1 kg of H₂O at 2 MPa is a saturated vapor. The water is slowly compressed in an isobaric process until it is entirely saturated liquid.
- Sketch the process on P-v and T- v plots. Label the initial and final states.
 - Determine the change in temperature of the fluid. Explain your answer.
 - Determine the work interaction (J). Is the work “in” or “out” of the system?
 - Determine the heat interaction (J). Is the heat “in” or “out” of the system?
5. (20 points) Consider a piston-cylinder system with 1 kg of water that undergoes the cycle in the table below. Processes 1-2 and 3-4 are adiabatic. Calculate the work and heat transfer for each of the processes, and the net work and net heat transfer of the entire cycle. Feel free to use the NIST webbook to calculate properties you need if they are not present on the table. Discuss the result of the net work and net heat transfer – these numbers should look interesting to you when you calculate them.

State	P [MPa]	T [K]	v [m ³ /kg]
1	2.6	499.2	0.0012014
2	5	500	0.0012
3	5	578.5	0.046114
4	2.6	499.2	0.076899